

L09 Introduction to Mechanism Design

CS 280 Algorithmic Game Theory
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Inspired by
by J. Hartline and T. Roughgarden notes

Warm-up: Single-item allocation

Definition (Single-item). *The single-item allocation problem is given by*

- *a single **indivisible** item,*
- *n agents **competing** for the item,*
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- Ask the agents/bidders to **report their values (bids)**, each agent reports b_i .
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- **Allocate** item to i^* .

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Outcome of mechanism is unpredictable, hard to reason about performance

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Proof. Consider $v_1 = 1$ and $v_i = \epsilon$ for $i \geq 2$. Expected surplus is

$$\frac{1}{n} (1 + (n-1)\epsilon) .$$

Thus approximation ratio is $\frac{n}{1+(n-1)\epsilon} \rightarrow n$ as $\epsilon \rightarrow 0$.

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Charge payments, proportionally to the agents' bids => Discourage low-valued agents from making high bids.

Single-item auctions

Approach 2 (First-price auction). *The first-price auction is defined:*

- *Agents report their **bids** b_i .*
- *Select agent $i^* = \arg \max_i b_i$ (**highest bid**).*
- *i^* gets the item and pays the amount of b_{i^*} .*

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First-price auctions are **hard to reason** about.
As a participant, it's hard to figure out **how to bid**.
As an auction designer, it's **hard to predict** what
will happen.

Single-item auctions

Approach 3 (**Second-price auction**). *The second-price or Vickrey auction is defined:*

- *Agents report their **bids** b_i .*
- *Let agent $i^* = \arg \max_i b_i$ and let j^* be the agent with (**second**) highest bid.*
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Theorem (Vickrey is truthful). *In second price auctions, every bidder i has a dominant strategy: Set her bid b_i equal to her private v_i (report truthfully). Dominant means the utility of bidder i is maximized no matter what other bidders do.*

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Remark: Utility of bidder i is $u_i := v_i - p_i$ **if he gets the item** and $u_i := 0$ **otherwise**.

Second price auctions

Proof. Fix an agent i and set $B = \max_{j \neq i} b_j$.

Consider the cases:

- If $b_i < B$ then i gets utility 0.
- If $b_i \geq B$ then i wins the item and $u_i = v_i - B$.

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Assume that $v_i < B$.

- If i reports truthfully, she gets utility 0 (did not win the item).
- Assume not and $b_i < v_i$ then the utility of i will still be 0.
- Assume not and $B > b_i > v_i$ then the utility will still be 0.
- Assume not and $b_i \geq B > v_i$ then the utility will be negative.

Second price auctions

Proof cont. Fix an agent i and set $B = \max_{j \neq i} b_j$.

Assume that $v_i \geq B$.

- If i reports truthfully, she gets utility $v_i - B \geq 0$ (won the item).
- Assume not and $b_i > u_i$ then the utility of i will still be $v_i - B \geq 0$.
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No matter what other bidders do, truthtelling is best strategy (**Dominant strategy**).

A general approach

An auction should satisfy following properties:

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- (Computational Efficiency) The auction can be implemented in **polynomial time**.

An Example

Problem: Consider a society of n citizens and public good G .

- Each agent has (private) valuation v_i for the good.
- Cost of building G is (publicly known) C .
- G should be built if $\sum_{i=1}^n v_i > C$.

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Answer: For citizen i , if $v_i > \frac{C}{n}$, i should report $C + \epsilon$ so G will be built!

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Answer: **No DSIC!** $+ \epsilon$ so G will be built!

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Solution: Charge citizen i the amount $p_i := \max(0, C - \sum_{j \neq i} v_j)$.
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Definition (**Quasi-linear environments**). Also known as Vickrey-Groves-Clark (**VCG**) environments:

- n agents,
- Set of outcomes (finite) $\mathcal{X}$,
- Each agent i has a valuation $v_i : \mathcal{X} \rightarrow \mathbb{R}^+$,
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Remark: This framework is called mechanism design **with money**.

VCG mechanisms

Definition (VCG mechanism). *The family of mechanisms is defined as follows:*

- *Agents have valuations v_i and report their bids b_i .*
- *Set $x^* = \arg \max_{x \in \mathcal{X}} \sum_{i=1}^n b_i(x)$.*
- *Each agent pays $p_i := \underbrace{h_i(b_{-i})}_{\text{without } i} - \underbrace{\sum_{j \neq i} b_j(x^*)}_{\text{with } i}$*
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Theorem (VCG is DSIC). *Every VCG mechanism is DSIC.*

Proof. Fix an agent i and let $x^* = \arg \max \sum_{i=1}^n v_i(x)$. Assume that i reports $b_i \neq v_i$ and x' be the maximum if i reports b_i .

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By definition of x^* $u_i \geq u'_i$

- Each agent has utility $u_i = v_i(x^*) - p_i(x^*)$.

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Proof. Fix an agent i and let $x^* = \arg \max \sum_{i=1}^n v_i(x)$. Assume that i reports $b_i \neq v_i$ and x' be the maximum if i reports b_i . Observe that

$$\begin{aligned} u_i &= v_i(x^*) + \sum_{j \neq i} v_j(x^*) - h_i(v_{-i}) \text{ if } i \text{ reports } v_i \text{ and} \\ u'_i &= v_i(x') + \sum_{j \neq i} v_j(x') - h_i(v_{-i}) \text{ if } i \text{ reports } b_i. \end{aligned}$$

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Remark: For single-item, this is the **second-price** auction!

VCG might not be efficiently computable (e.g., **combinatorial auctions**)